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Nanoceria Supported Single-Atom Platinum Catalysts for Direct Methane Conversion

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Supporting Information Placeholder

ABSTRACT: Nanoceria-supported atomic Pt catalysts (denoted as Pt₁@CeO₂) have been synthesized and demonstrated with advanced catalytic performance for the non-oxidative, direct conversion of methane. These catalysts were synthesized by calcination of Pt-impregnated porous ceria nanoparticles at high temperature (ca. 1,000 °C), with the atomic dispersion of Pt characterized by combining aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM), X-ray photoelectron spectroscopy (XPS), X-ray absorption spectroscopy (XAS) and diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) analyses. The Pt₁@CeO₂ catalysts exhibited much superior catalytic performance to its nanoparticulated counterpart, achieving 14.4% of methane conversion at 975 °C and 74.6% selectivity toward C₂ products (ethane, ethylene and acetylene). Comparative studies of the Pt₁@CeO₂ catalysts with different loadings as well as the nanoparticulated counterpart reveal the single-atom Pt to be the active sites for selective conversion of methane into C₂ hydrocarbons.

Introduction

Over the recent years, natural gas has risen as a clean and cost-effective source of hydrocarbons, with great potential for replacing coal and crude oil in many sectors of energy and chemical industries.¹⁻² The conventional approaches for methane conversion via syngas (a mixture of CO and H₂) is however challenged by the low carbon efficiency, large loss of exergy and high capital cost associated with the complex, multistage processes.³⁻⁵ Alternatively, direct conversion of methane can be achieved via oxidative coupling⁶⁻⁷ or non-oxidative dehydrogenation⁸⁻¹⁰ to produce olefins or aromatics. These approaches are believed to be more economical and environment-friendly than via the syngas.¹¹⁻¹²

Single atom catalysts (SACs) represent a new frontier of heterogeneous catalysis and have been extensively studied

for many reactions, including CO oxidation,¹³⁻¹⁴ water-gas shift,¹⁵⁻¹⁶ methane steam reforming,¹⁷ selective hydrogenation of alkynes and dienes¹⁸⁻¹⁹ and so on. The superior catalytic performance are usually attributed to the atomic dispersion of metal atoms with low coordination number, quantum confinement and/or strong metal-support (mostly metal oxides) interactions.²⁰⁻²² It has also been reported that atomic Fe sites embedded in a silica matrix give rise to high catalytic selectivity for the non-oxidative conversion of methane to ethylene, aromatics and hydrogen; the absence of metal ensembles suppresses C-C coupling and carbon coking, giving rise to long-term stability under the high-temperature reaction conditions.¹⁰ Despite the progress, it remains elusive in many cases whether the single-atom sites offer distinct catalytic mechanisms from their ensembled counterparts with continuous surfaces.²¹⁻²² One main obstacle is the lack of robust control over the dispersion of metals during the synthesis, and sub-nanometer clusters or nanoscale metal particles usually co-exist with single metal atoms in the composite materials. The other challenge is the stability of the SACs under reaction conditions, especially considering the potential atomic aggregation and agglomeration at high temperatures.

Here we report the synthesis of ceria (CeO₂)-supported atomic Pt catalysts for direct conversion of methane into light hydrocarbons. Pt has been widely used to activate the C-H bond in hydrocarbons,²³⁻²⁵ but carbon coking usually takes place on the conventional catalysts composed of Pt clusters or nanoparticles at high temperatures (e.g., >800 °C), which has limited the application of Pt-based catalysts for methane conversion²⁶⁻²⁷. In this study, nanoceria-supported atomic Pt catalysts were synthesized by calcination of Pt-impregnated porous CeO₂ nanoparticles (Figure S1) at ca. 1,000 °C (see the Supporting Information for experimental details). The obtained Pt₁@CeO₂ catalysts were characterized by using HAADF-STEM and XPS, and the absence of Pt ensembles was further confirmed by DRIFTS analysis using CO as a molecular probe. The Pt₁@CeO₂ catalysts of various loadings

(with 0.5 – 1.0 wt% of Pt) were then evaluated for the methane conversion reaction, and the catalytic performance was further compared to their nanoparticulated counterpart to reveal the single-atom active sites.

Results and Discussion

Figure 1a shows the representative transmission electron microscopy (TEM) image of the Pt_t@CeO₂ catalyst with ~0.5 wt% of Pt. They exhibit a slab-like morphology with the size varying from ~15 to ~40 nm, which were likely transformed from the porous nanospheres. High-resolution HAADF-STEM images reveal that Pt is dispersed on the CeO₂ nanoslabs at the atomic scale (Figures 1 b-d). In these images, individual Pt atoms are exhibited as bright dots with higher contrast than the surrounding CeO₂ lattice (Figures 1 e, f). The slab-like nanocrystals exhibit lattice fringes with the spacing measured to be ca. 0.31 nm, which can be assigned to the (111) planes of CeO₂ in the fluorite phase (Figure 1b). Further synthesis shows that the ratio of Pt can be tuned from ca. 0.25 – 1.0 wt%, although Pt clusters start to appear in the 1.0% Pt_t@CeO₂ (Figure S2). For comparison, 3 nm Pt nanoparticles were also synthesized and deposited on similar CeO₂ nanoslabs (with 0.5 wt% of Pt, denoted as PtNPs/CeO₂) (Figures S3 and S4). X-ray diffraction (XRD) patterns collected for the Pt_t@CeO₂ catalyst only show the CeO₂ peaks in the fluorite (Fm $\bar{3}$ m) phase (Figure S5), where the absence of Pt-phase peaks is consistent with the atomic dispersion of Pt as observed in the STEM images. The single-atom dispersion of Pt in the catalysts was further confirmed by CO chemisorption (Table S2) and extended X-ray absorption fine structure (EXAFS) analyses (Figure S8a). In particular, the EXAFS spectrum for 0.5% Pt_t@CeO₂ only exhibits one pronounced peak associated with the first-shell Pt-O bond (~2.03 Å), where the absence of Pt-Pt and higher-shell Pt-O-Pt scattering demonstrates atomic dispersion of Pt in the catalyst.

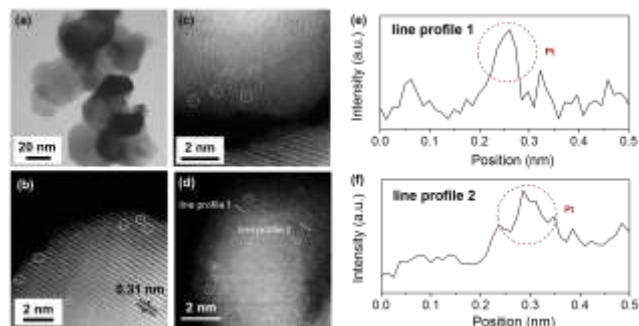


Figure 1. Representative (a) TEM and (b-d) high-resolution HAADF-STEM images of the Pt_t@CeO₂ catalyst with 0.5 wt% of Pt. (e, f) Intensity profiles of the scans along the dash lines marked in (d).

Oxidation state of Pt in the catalysts was characterized by using XPS (Figure 2). The spectrum collected for the 0.5% Pt_t@CeO₂ catalyst shows two peaks at the Pt 4f edge with binding energies of 73.7 eV and 76.9 eV, which are assigned to the 4f_{7/2} and 4f_{5/2} states of Pt²⁺, respectively (Figure 2a).²⁸⁻²⁹ For PtNPs/CeO₂, the Pt 4f doublet exhibits downshift by ~1 eV in binding energy (Figure 2c). Deconvolution analysis reveals the presence of two additional peaks at 72.2 and 75.7 eV, in addition to the aforementioned two peaks associated with Pt²⁺, which can be assigned to the same spin-orbital split of metallic Pt (Pt⁰). The atomic ratio between Pt²⁺ and

Pt⁰ was estimated to be ~1.5 in the PtNPs/CeO₂ catalyst, with the oxidized Pt likely coming from surface oxidation of the Pt nanoparticles during calcination (see the Supporting Information for the details of synthesis).

The XPS spectra collected at the Ce 3d edge are shown in Figures 2 b and d. The spectra can be deconvoluted on the basis of two multiplets that correspond to the 3d_{3/2} and 3d_{5/2} core holes of Ce (denoted as u and v, respectively) and have a spin-orbit splitting of ~18.6 eV.³⁰⁻³¹ u^o (898 eV) and v^o (880 eV) for Ce(3d⁹4f¹)-O(2p⁶), uⁱ (901 eV) and vⁱ (882 eV) for Ce(3d⁹4f²)-O(2p⁴), uⁱⁱ (906 eV) and vⁱⁱ (889 eV) for Ce(3d⁹4f³)-O(2p⁵) and uⁱⁱⁱ (916 eV) and vⁱⁱⁱ (897 eV) for Ce(3d⁹4f⁰)-O(2p⁶). The states marked with u^o/v^o and uⁱ/vⁱ are features of Ce³⁺, which was estimated to occupy ~46% and 33% of the Ce species in the 0.5% Pt_t@CeO₂ and PtNPs/CeO₂ catalysts, respectively (Table S1). These results indicate that the CeO₂ nanoslabs employed as support here are rich in Ce³⁺ defects and oxygen vacancies, which is likely a result of oxygen evolution during the high-temperature (1000 °C) treatment.

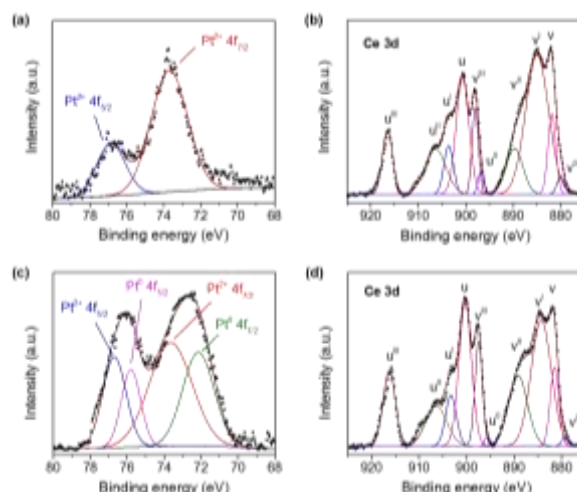


Figure 2. XPS spectra collected on the (a, b) 0.5% Pt_t@CeO₂ and (c, d) PtNPs/CeO₂ catalysts at the (a, c) Pt 4f and (b, d) Ce 3d edges.

The XPS analysis shows that, in the Pt_t@CeO₂ catalysts, Pt was dispersed on the CeO₂ support in the oxidized form (Pt²⁺). It was reported that Pt can be emitted as volatile PtO_x above 800 °C in air.¹⁴ The porous nanospheres (Figure S1) employed as precursor in the present study could have helped trap the PtO_x vapor, while Ce(III) and oxygen vacancies enriched on the formed CeO₂ nanoslabs represent coordinatively unsaturated, electrophilic sites and could have attracted and stabilized the atomic platinum oxides, e.g., in the form of planar Pt²⁺O₄ clusters³². Thereby Pt was favorably dispersed as single-atom species in the derived Pt_t@CeO₂ catalysts.

To gain a more extensive evaluation of the atomic dispersion of Pt, we have further performed diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) analysis of CO adsorption on the Pt_t@CeO₂ catalysts. This method has previously been demonstrated to be effective in identification of single Pt atoms on oxide supports.¹³ Figure 3 compares the absorption spectra recorded on the Pt_t@CeO₂ catalysts with various loadings of Pt (0.25, 0.5 and 1.0 wt%) and PtNPs/CeO₂ (0.5 wt%), where CO was pre-adsorbed at different partial pressures. Only one peak was observed at 2,089

cm^{-1} for the Pt_i/CeO_2 catalysts with 0.25% and 0.5% of Pt (Figures 3 a, b), which can be assigned to the linearly bonded CO (CO_L) on $\text{Pt}^{\delta+}$.¹³ An additional peak at $1,991\text{ cm}^{-1}$ appears in 1.0% Pt_i/CeO_2 (Figure 3c), as well as $\text{PtNPs}/\text{CeO}_2$ (Figure 3d), which can be ascribed to the bridge bonded CO (CO_B) on Pt, a typical feature of Pt ensembles with continual surfaces.^{13, 33} $\text{PtNPs}/\text{CeO}_2$ exhibits another peak at $2,078\text{ cm}^{-1}$ in addition to the two peaks observed on 1.0% Pt_i/CeO_2 , which could be assigned to CO_L on the Pt nanoparticles with, for example, metallic Pt sites.³⁴ The absence of the CO_B peak thereby confirms the isolation of Pt sites in the Pt_i/CeO_2 catalysts at relatively low Pt ratios (e.g., <1%), whereas Pt clusters have formed in the case with higher loadings.

After demonstrating the atomic dispersion of Pt, the Pt_i/CeO_2 catalysts were further evaluated for nonoxidative conversion of methane at $900 - 1,000^\circ\text{C}$ with a space velocity of $6\text{ L}/(\text{g}_{\text{cat}}\cdot\text{h})$. Figure 4a summarizes the temperature-dependent methane conversion and product distribution for the 0.5% Pt_i/CeO_2 catalyst. The methane conversion increased with temperature and reached 23.1% at $1,000^\circ\text{C}$. The selectivity of C_2 hydrocarbons exhibited gradual decrease from 98.4% at 900°C to 66.7% at $1,000^\circ\text{C}$. The amount of C_3 product was rather small and always <10% throughout the investigated temperature range. At temperatures $\geq 950^\circ\text{C}$, aromatic products started to appear and the selectivities increased with temperature, achieving 26.6% for benzene and 2.1% for naphthalene at $1,000^\circ\text{C}$. Breakdown of the C_2 product distributions is further elucidated in Figure 4b. At relative low temperatures, ethylene and ethane were the two dominant products, with the selectivity measured to be 51.1% and 43.6% at 900°C , respectively. At elevated temperatures, acetylene became more abundant and its selectivity achieved 41.7% at $1,000^\circ\text{C}$, whereas only 19.8% of ethylene and 5.1% of ethane were left at this temperature. It is noticed that the amount of hydrogen generated from the methane conversion matches well with the concentrations calculated from the reaction stoichiometries and mass balance by taking the various hydrocarbon products into account (Figure S9).

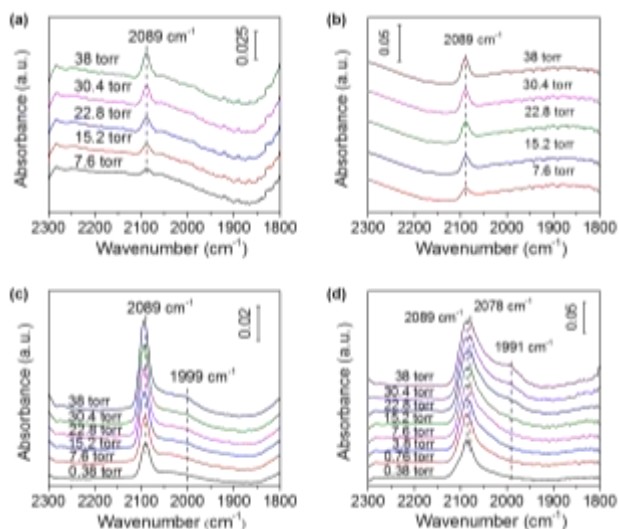


Figure 3. DRIFTS of CO chemisorption at different CO partial pressures on Pt_i/CeO_2 with various weight percentages of Pt: (a) 0.25%, (b) 0.5% and (c) 1.0%. (d) $\text{PtNPs}/\text{CeO}_2$ (0.5 wt%) was also shown for comparison.

The atomic Pt catalyst is much superior to its nanoparticulated counterpart for the methane conversion reaction.

Figure 4c provides the comparison of methane conversion and product selectivity for the three Pt_i/CeO_2 catalysts with different Pt loadings as well as the $\text{PtNPs}/\text{CeO}_2$ catalysts at 975°C (see the calculated turnover frequencies (TOFs) in Table S2). It is noted that even in the blank reaction tube, CH_4 had a conversion of 1.1% at this temperature due to the non-catalytic, thermal activation and dehydrogenation, but no hydrocarbons were detected at significant amounts (albeit with some ethane at 3.2% selectivity), suggesting that the converted methane mostly became coke (Figure S10) and deposited on the tube wall. The bare CeO_2 support exhibited somewhat higher (6.9%) CH_4 conversion, but coke was still the dominant (88.3%) product. The $\text{PtNPs}/\text{CeO}_2$ catalyst had a CH_4 conversion of 9.7%, with 8.3% and 6.3% selectivities toward ethylene and acetylene, respectively, whereas the majority (79.8%) of carbon also ended up in coke deposition. All of the three atomic Pt catalysts performed higher activities than their nanoparticulated counterpart, achieving >12% conversion of methane and >95% of total selectivity toward hydrocarbons. Among the three Pt_i/CeO_2 catalysts, the 0.5% one demonstrates the highest conversion (14.4%) of methane and selectivity (74.3%) toward C_2 products, where ethylene (33.2%) and acetylene (35.1%) represent the major products. The slightly lower methane conversion and hydrocarbon selectivity from 1.0% Pt_i/CeO_2 can be ascribed to the presence of a small amount of Pt clusters in this catalyst, as can be seen from the STEM images (Figure S2) and DRIFTS analysis (Figure 3c).

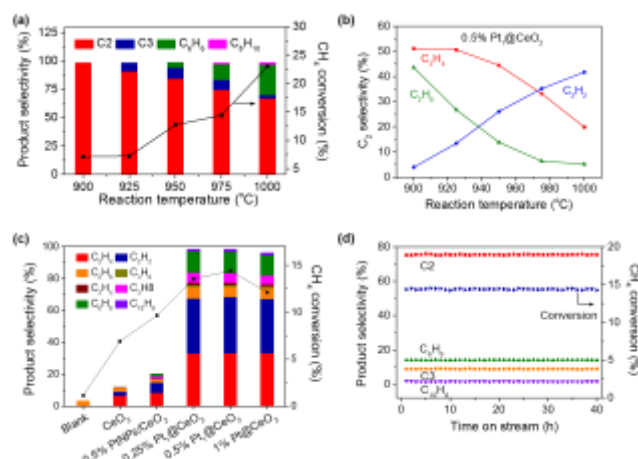


Figure 4. Catalytic performance for the nonoxidative CH_4 conversion evaluated at $6\text{ L}/(\text{g}_{\text{cat}}\cdot\text{h})$. (a) Catalytic activities and selectivities of the 0.5% Pt_i/CeO_2 catalyst as functions of the reaction temperature. Black squares represent CH_4 conversion, and the colored histograms for product distributions. Here the light hydrocarbons are categorized as C_2 (ethane, ethylene and acetylene) and C_3 (propane, propylene and propyne) hydrocarbons, with further breakdown of the C_2 products shown in (b). (c) Comparison of methane conversion and product distributions at 975°C over the different catalysts and the controls. (d) Stability test of the 0.5% Pt_i/CeO_2 catalyst performed at 975°C .

The above observations clearly point to the single-atom Pt as active sites, which are not only able to activate the C-H bond by dehydrogenation, but also direct the C-C coupling toward favorable formation of C_2 hydrocarbons. In the cases of Pt nanoparticles and clusters, although the continuous metal surface is also activate for dehydrogenation, it lacks

control over the extent of C-C coupling and cannot suppress the coking, which is consistent with the reported behavior of conventional catalysts with ensembles of metal atoms.^{26-27, 35}

Besides suppression of carbon coking, the performance of the Pt₁@CeO₂ catalysts is noticeably different from the previously reported atomic Fe@SiO₂ catalysts, albeit with similar methane conversions (e.g., 12.7% for Pt₁@CeO₂ versus ~8% for Fe@SiO₂ at 950 °C).¹⁰ The single-atom Pt catalysts reported here give rise to much higher C₂ product selectivity than the Fe@SiO₂, e.g., ~84% versus ~47% at 950 °C. In the latter case, the rest products are mainly aromatics (consistently at ~50% independent of the reaction temperature) and nearly equally distributed between benzene and naphthalene. While the Pt₁@CeO₂ catalysts produce all the three kinds of C₂ species, ethylene is the only C₂ product obtained with the Fe@SiO₂ catalyst. These differences suggest that the Pt₁@CeO₂ catalysts may possess distinct catalytic mechanisms, particularly in the C-C coupling steps, from the Fe@SiO₂ catalyst, where multi-carbon species were believed to form from gas-phase methyl ($\cdot\text{CH}_3$) radicals via noncatalytic thermochemical processes.¹⁰

We have performed thermodynamic calculations for the equilibria of the involved conversion reactions toward different products (see the Supporting Information, section 3). The ratio of C₂ species in the product was found to be much higher than at equilibrium (Table S4). DRIFTS analysis of the Pt₁@CeO₂ catalyst after methane activation at 900 °C revealed the presence of π -bonded ethylene and acetylene, suggesting that the single Pt sites may be capable of stabilizing C₂ adsorbates (Figure S11). We thus propose that the methane conversion on the Pt₁@CeO₂ catalysts involves adsorbing C₂ intermediates, possibly formed from catalytic coupling of two dehydrogenated C₁ adsorbates (such as *CH₃ and *CH₂) on the single Pt sites (other than adjacent Pt sites as on the surface of Pt ensembles). As further coupling to form C₃ adsorbates may be inhibited on the atomically dispersed Pt, these C₂ intermediates must desorb and thus favor the production of C₂ hydrocarbons. The minor C₃ and aromatic products obtained at higher temperatures can be ascribed to the thermochemical dehydrogenation and oligomerization of the C₂ molecules in gas phase.

As discussed above, the primary role of the CeO₂ support in the Pt₁@CeO₂ catalysts is the stabilization of the single-atom Pt species, which could also be true under the reaction conditions.^{32, 36} However, synergetic effects may also be present between CeO₂ and Pt on the activation of CH₄. Active sites have been shown to be present at the interface between Pd clusters and CeO₂ with low methane activation barriers.³⁷ Although the situation may be different in the single-atom catalysts reported here, it is noticed that the onset temperature of methane activation on Pt₁@CeO₂, i.e., ≤ 900 °C, is lower than that (≤ 950 °C) previously reported for single-atom Fe@SiO₂.¹⁰ This is indicative of the active role of CeO₂ in the methane conversion process other than only being an inert support.

Ultimately, we demonstrate that both the high catalytic activity and selectivity of the single-atom Pt₁@CeO₂ catalysts are stable, with indiscernible drop after 40 h of reaction at 975 °C (Figure 4d). The high durability further confirms the suppression of carbon coking on the active sites. The single-atom platinum-ceria catalysts reported here thus possess great potential for practical implementations.

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version for submission.

Notes

The authors declare no competing financial interests.

ASSOCIATED CONTENT

Supporting Information

More experimental details, characterizations and catalytic results. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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SYNOPSIS TOC

